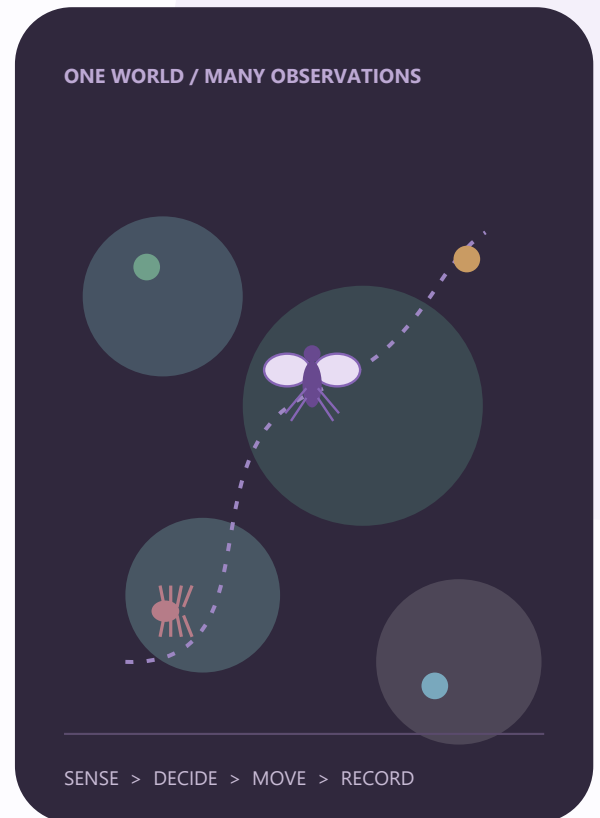


BRAIN RESEARCH, MADE EXPLORABLE

MOSK

A small life, unfolding.

What happens when a published brain map becomes a question you can test on screen? MOSK is an interactive observatory: follow one virtual fruit fly, change its world, and inspect the signals behind each visible response.



THE PRODUCT IN THREE MOVES

01 OBSERVE

Watch the fly explore a world that unfolds around it.

02 INTERVENE

Place food, water or a spider and compare what changes.

03 EXPLAIN

Read controller signals, life events and real source annotations.

A clear scientific boundary. The fly on screen is an authored simulation. Google Research and collaborators published anatomical wiring data, not a ready-made virtual animal. MOSK makes that research approachable while preserving the distinction.

The research beneath the experience

Three related layers serve different purposes. Their provenance is visible throughout MOSK.

PUBLISHED / MALECNS

Google and collaborators mapped the male fly's central nervous system.

The Cell study describes **166,691 neurons** across brain and ventral nerve cord, with roughly **125 million synaptic contacts**. The released structural map supports tracing sensory-to-motor paths and comparing cell types; it is not a recording of a living fly's activity.

IN MOSK / SOURCE ATLAS

A real anatomical sample you can inspect.

The MaleCNS atlas contains **96 source-identified annotation records** with cell labels, classes and provenance. Visitors can search and inspect them. It does not bundle the full connection graph or make those cells control the roaming fly.

THREE LAYERS, CLEARLY LABELED

01

Anatomy

Real MaleCNS identifiers and cell annotations.

02

Roaming

A designed, interpretable 18-channel controller.

03

Memory

A separate published mushroom-body assay.

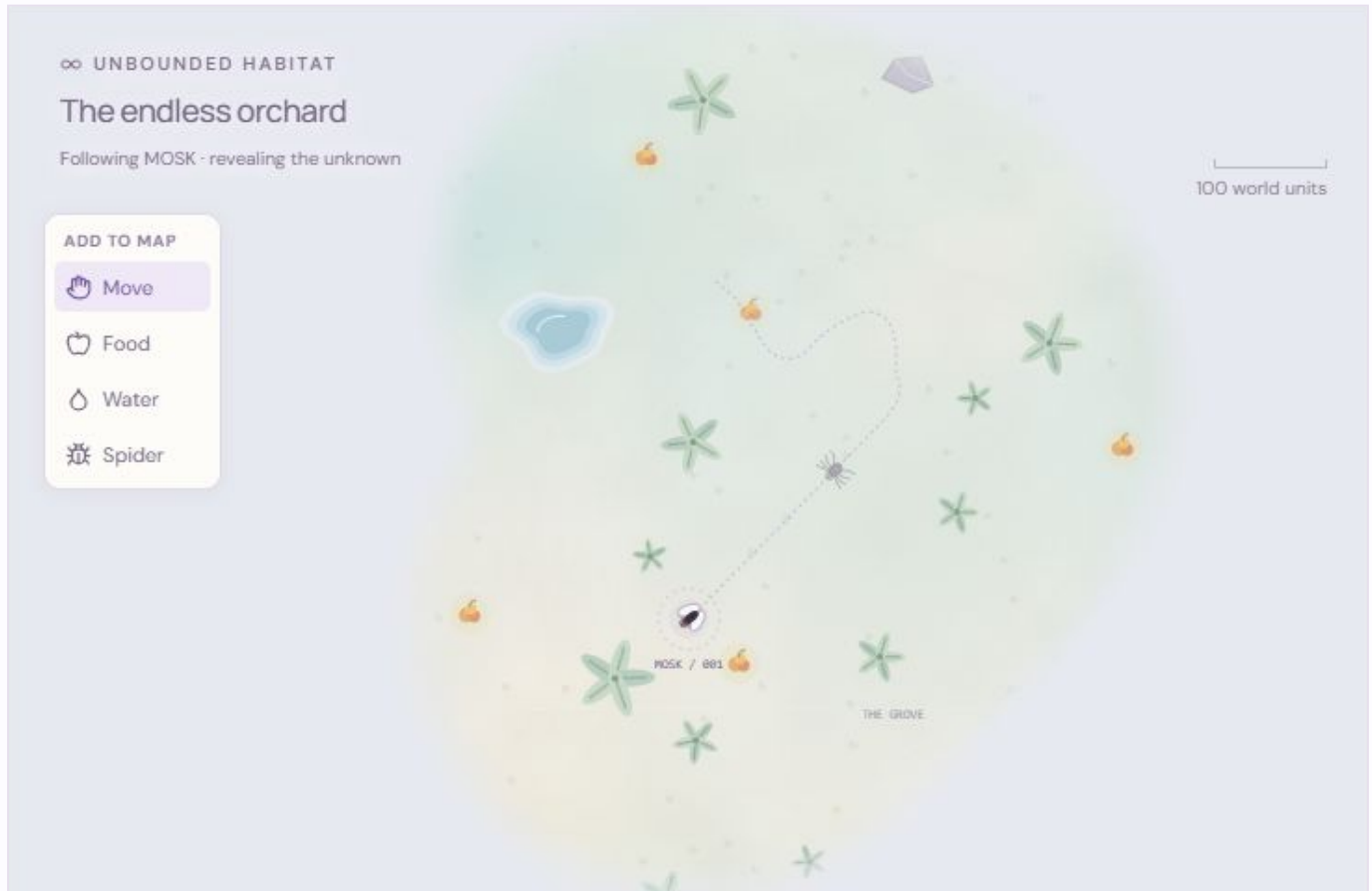
MEMORY ASSAY / INDEPENDENT STUDY

A second published model makes learning observable without pretending that the MaleCNS map already supplies it.

The Research view runs the Huang-Luo Nature 2024 mushroom-body model: odor conditioning, retention and extinction change modeled synaptic weights and circuit response rates. Its checkpoint belongs to a life record, but those weights **do not steer the roaming fly**. The assay is not part of Google's MaleCNS release.

A world that invites experiments

A real MOSK session: the map, intervention brush and visible trail during a spider encounter.



ACTUAL INTERFACE / The fly, a placed spider, resources, cover and fogged terrain.

01 / THE LIVING WORLD

Explore only what the fly reaches.

Organic terrain is generated around the animal rather than all at once. Its movement reveals new sectors, while food and water are consumed and replenished. Spiders can patrol and pursue; cover can break detection.

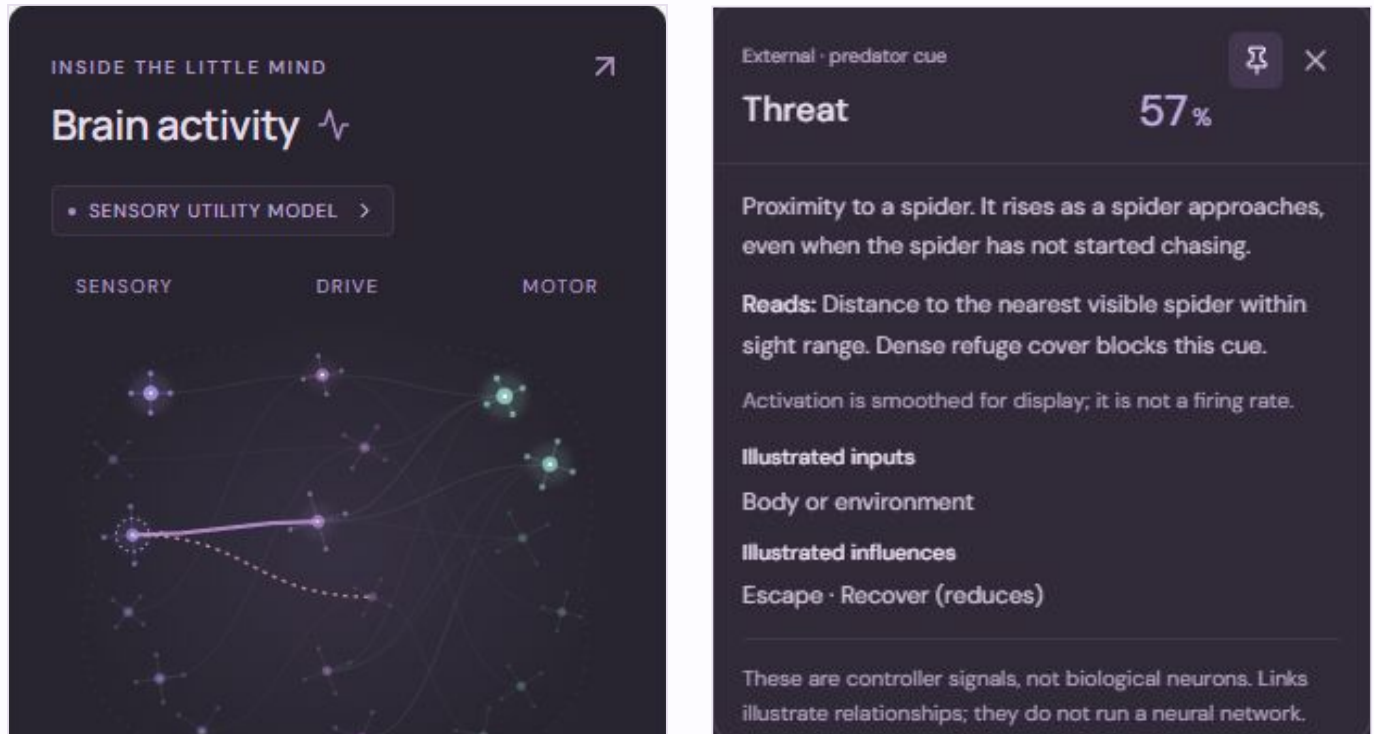
02 / CHANGE ONE CUE

Place, pause and compare.

Paint food, water or a spider onto explored terrain. Follow or pan the camera, zoom with the mouse wheel, and choose 1x, 2x or 5x playback. Energy, hunger, hydration, fatigue, health and arousal show the consequences of each choice.

Follow the response, then the record

The controller view explains a cue; the journal preserves the event in this life.



BRAIN ACTIVITY / Eighteen designed channels. Pinned Threat shows its input, level and illustrated influence - not measured neuron firing.

03 / EXPERIMENT JOURNAL

Field notes	6 events
Entered new terrain: sector O,1.	00:00:18
Observer placed a spider at (630, 639). Another spider owns the current chase; this one roams for now.	00:00:17
Predator detected. Escape response engaged.	00:00:13
Observer placed a spider at (861, 396). It detected the fly and will pursue when the simulation runs.	00:00:13
Found a food source. Feeding.	00:00:01
Generation 001 entered the habitat.	00:00:00

ACTUAL FIELD NOTES / Interventions, encounters and terrain discoveries receive timestamps; charts and archived lives support comparison.

Learning with visible limits

A separate published memory assay, real atlas annotations and a reproducible local run.

What is stored right now

Δ effective KC $\rightarrow$ MBON weight

Six values hold the acquired association: one per odor and compartment. Zero means the initial fitted connection. Negative values mean depression. These are model weight units, not a count of synapses or a confidence score.

Odor	y1 - short term	$\alpha 2$	$\alpha 3$
CS+ - paired odor	-15.786	+10.949	+4.934
CS- - comparison	+8.248	+11.354	+16.144

Odor responsiveness: CS+ 46.4%, CS- 34.1%. This separate adaptation state changes with exposure and recovers during gaps.

MEMORY ASSAY / After eight published-protocol bouts, the model's six odor-specific connection weights differ from their initial state.

04 / MEMORY ASSAY

The Huang-Luo mushroom-body model steps through odor conditioning, retention and extinction. Its weights and live log are saved with this life, but do not steer the roaming fly.

05 / MALE CNS ATLAS

Search 96 source-identified MaleCNS annotation records. These are real anatomical labels, not the controller's 18 displayed channels or live neuron activity.

TECHNICAL FOUNDATION

React + TypeScript + Vite host a seeded, deterministic 30 Hz local simulation. Terrain generates near the fly; local cues feed an interpretable utility controller. IndexedDB autosaves lives and assay checkpoints, while JSON import/export makes portable backups. New generations archive the previous life without inherited learning.

Explore the demo and original sources

Live MOSK demo - moskdemo.netlify.app

Google Research overview of the MaleCNS release

Berg et al., Cell 2026 - doi.org/10.1016/j.cell.2026.08.015

Official dataset and provenance - male-cns.janelia.org

Huang, Luo et al., Nature 2024 - doi.org/10.1038/s41586-024-07819-w

MOSK source code - github.com/francopignanelli/mosk

Independent project by **Franco Pignanelli**. MOSK is not affiliated with Google or HHMI Janelia.